

Huaijin Ran

+86 156-6096-1639 | Seventeen17510@gmail.com
GitHub | Google Scholar | LinkedIn

Education

Nanyang Technological University (NTU), Singapore
MSc, Cyber Security

Aug. 2026 – Jul. 2027 (Expected)

University of Liverpool, United Kingdom
BSc (Hons), Intelligent Supply Chain

Sep. 2022 – Jul. 2026

Final-Year GPA: **3.65/4.00** | Major GPA: **3.51/4.00**

Publications

SentryLine: Reviewable Capability Traces for CPython Plugin Hosts Jul 2026
(Second Author, Accepted to SPLASH/ISSTA 2026 Posters Track)

Co-developed an admission-time review system that combines interpreter-side observations with wrapper-level guards to produce auditable capability traces for CPython plugins; evaluation preserved all 28 benign workloads and blocked all 72 misuse workloads across 13,888 boundary decisions.

Exploring Large Language Models for Compliance Violation Detection in Android Apps Jan 2026
(First Author, **Invited** to ICSE 2026 Posters Track)

Invited from Technical Track. First systematic evaluation of LLMs for privacy compliance detection across four regulations (GDPR, LGPD, PDPA, PIPEDA), revealing granularity-dependent model specialization and 2–3× cross-regulation performance disparities.

Who Wrote This Patch? Toward Accountable Automated Program Repair Mar 2026
(First Author, Accepted to FSE 2026 IVR Track)

Proposed multi-patch fingerprinting via behavioral signatures to address the accountability vacuum in LLM-generated Automated Program Repair (APR).

MSESTA: Multi-Scale Evolving Spatial–Temporal Attention for Long-Term Traffic Forecasting Aug 2025
(First Author, Accepted to Neural Networks)

Proposed a novel multi-scale evolving spatial–temporal attention mechanism for traffic forecasting, optimized for memory efficiency and low latency.

ConstantGuard: Lattice-Based Taint Analysis for Timing Side-Channel Risk Detection in C Code Feb 2026

(Second Author, Accepted to FSE 2026 Posters Track)
Developed a static analyzer combining a three-level taint lattice and semantic checks for timing side-channel risks, reaching 92.86% function-level accuracy and exceeding 81,000 LOC throughput in under 32 seconds.

SPECTRA: Smart Contract Specification Inference via Abstract Interpretation Feb 2026
(Second Author, Accepted to NDSS 2026 Posters Track)

Proposed a novel framework that automatically infers Hoare logic specifications for smart contracts via abstract interpretation over an eight-domain product lattice. Achieved 99% success rate on 100 production contracts and identified 111 vulnerabilities.

The Watermark Paradox: How Provenance Verification Paves the Road to Camouflaged Backdoors Mar 2026

(Second Author, Accepted to FSE 2026 IVR Track)
Demonstrated that AI code watermarking and backdooring are structurally isomorphic, revealing how provenance verification inadvertently enables camouflaged backdoors.

TriMedLM: Advancing Three-Dimensional Medical Image Analysis with Multi-Modal LLM Nov 2024
(Corresponding Author, Published in *BIBM 2024*)

Proposed a multi-modal LLM framework for 3D medical image analysis to advance image-text retrieval, report generation, and visual question answering.

Dual-Stream Neural Architecture for Integrative Analysis of Large Multimodal Model in Mood Disorder Classification Nov 2024
(Corresponding Author, Published in *BIBM* 2024)
Developed a dual-stream neural architecture integrating functional and structural MRI data to enhance mood disorder diagnosis.

SPECTRA: Specification-Oriented Static Analysis for Smart Contract Security and Behavioral Contract Inference Feb 2026
(Third Author, Accepted to FSE 2026 Posters Track)
Proposed a static-analysis framework for Solidity contracts integrating symbolic execution, seven-domain abstract interpretation, and SMT validation. Evaluated across 220 contracts, successfully analyzing 98.18% and reporting 673 vulnerability findings.

WasmCert: A Reproducible Hybrid Security Analysis Prototype for WebAssembly Feb 2026
(Third Author, Accepted to FSE 2026 Posters Track)
Presented a hybrid prototype combining static token-level checks with source-level heuristics and runtime trap evidence. Achieved 86.48% detection on 355 vulnerability-oriented cases with 96.21% precision.

Journal Revision

GDPR-Bench-Android: A Benchmark for Evaluating LLMs in GDPR Compliance Detection Jul 2025
(First Author, Submitting to TOSEM, **Major Revision**)
Developed a benchmark framework to evaluate LLMs on GDPR compliance violations with 500+ annotated instances. [Code]

Intelligent Agents for Software Engineering: A Systematic Literature Review Jul 2025
(Second Author, Submitting to TOSEM, **Major Revision**)
Conducted a systematic review of intelligent agents in SE, developing a task-driven taxonomy and research agenda. [Code]

MMSA: Mamba based on Multi-Scale Attention for Traffic Flow Prediction Apr 2025
(Second Author, Submitting to Neural Networks, **Decision in Process**)
Proposed a multi-view spatial attention model for traffic prediction with real-time performance and reduced computational costs.

Research Contributions

FedFence: A Projection Calculus for Proof-Carrying CI/CD Trust Jul 2026
(Acknowledged Contributor; IEEE S&P 2027 Cycle 1 submission advanced to Round 2)

Project Experience

6th Oxford Machine Learning Summer School Aug 2025
Attended advanced machine learning lectures and workshops at the University of Oxford.

Recent Advances in Natural Language Processing (RANLP) 2025 Sep 2025
Invited to present research findings regarding LLM compliance detection from the GDPR-Bench project.

East China Normal University (*Product Manager Intern*) May 2024 – Dec 2024
Participated in software development and managed product requirements throughout the lifecycle.

Prudential UK (*Asset Allocation Intern*) Jul 2024 – Sep 2024
Studied and implemented practical asset allocation strategies in a professional financial environment.

Competitions & Achievements

- Kaggle: Detect AI Generated Text — Bronze medal, top 8% finish
- Codeforces Competitive Programming — Rating 1681, Expert level

Patents & Software Copyrights

- E-commerce Supply Chain Recommendation Software — Copyright No. 2024SR0915671
- Deep Learning-based Financial Document Automation (First Author) — Patent pending
- A Dynamic Traffic Obstacle Warning System (Second Author) — Patent pending